Project 2

Lecturer: Shuai Wang

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Requirements for submitting the assignment

- **Email address : seudatabaseqa@gmail.com**
- **Format of Email subject and File name:**

  StudentNumber-Name-Project-X

  example : *71121000-张三-Project-1.zip*

- **Including :**

  ① *report_A2.doc / report_A2.docx / report_A2.pdf*

  ② *university_A2.mdb / university_A2.accdb*

Deadline: Beijing time, 11.01, 23:59:59

Requirements for submitting the assignment

Note:

- **Name and submit the files according to the prescribed format!**
- **Include the required files (docx, accdb, pdf..). Each one is indispensable!**
- **Submit all the files in a compressed format (zip, rar…)!**
- **Ensure the accuracy of your name, student ID, and even the subject you are submitting for!!!**
- **Submit your report on time!! Late submissions will not be accepted now!**

Project 2

I. Tutorial with Project Examples

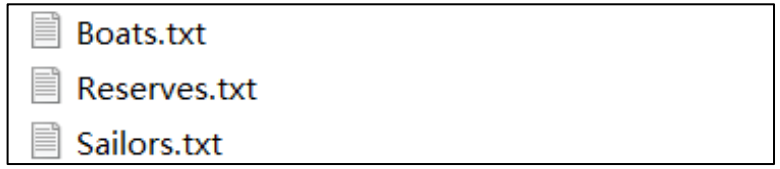
II. Assignment

Tutorial with Project Examples

1. Preparation Work

It is solely used for demonstrating the operational process and is not part of the experimental requirements.

①Download the following data files



②Import the txt into the lab2.accdb and create three tables

Sailor

sid	sname	rating	age
22	dustin	7	45
28	yuppy	9	35
31	lubber	8	55
32	Adam	7	30
34	Allen	6	40
35	Bart	8	50
37	Bill	7	30
39	black	6	35
41	white	7	45

Boats

bid	bname	color
101	tiger	red
103	lion	green
105	hero	blue
136	brave	gray
139	freedom	white
177	challenger	black
224	meteor	orange
269	rose	red
381	smart	blue

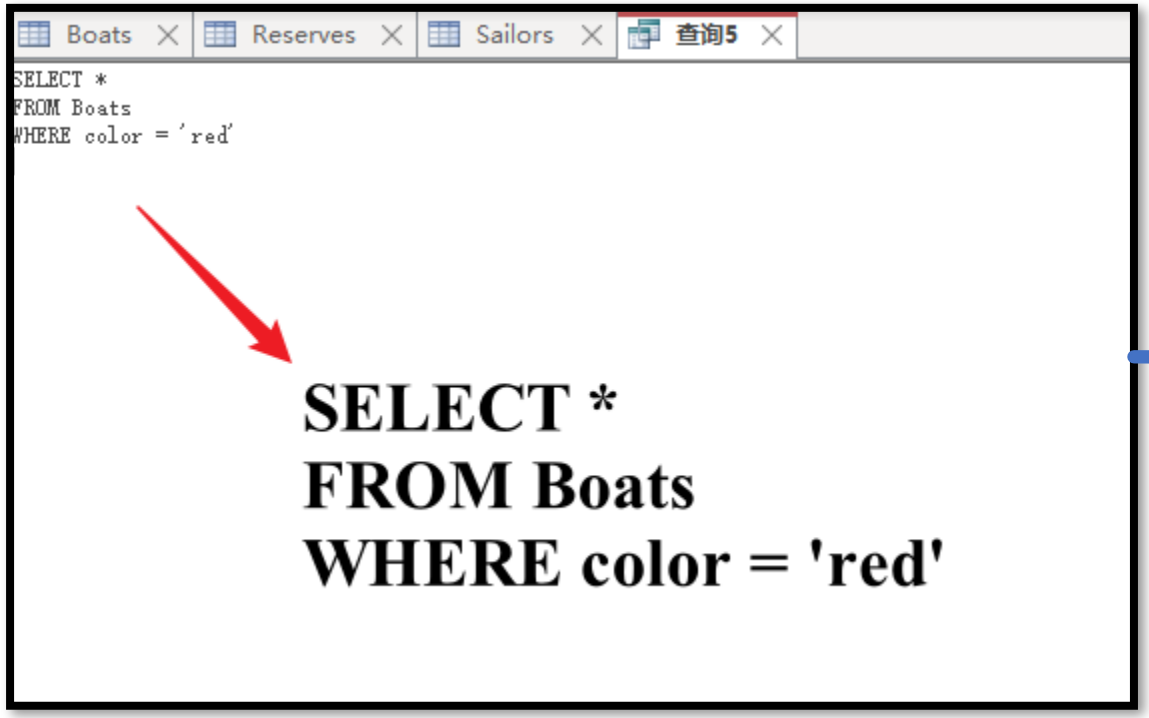
Reserves

sid	bid	day
22	101	2018/9/26
31	269	2019/10/6
32	101	2018/12/2
44	269	2019/10/14
47	177	2019/9/5
49	105	2019/4/13
51	224	2019/10/6
55	139	2019/8/12
58	103	2019/1/2
59	136	2019/7/9

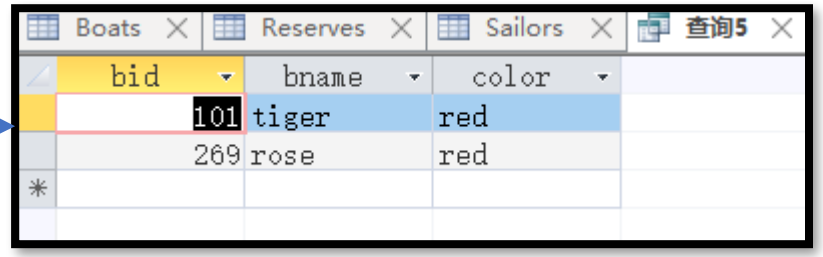
Tutorial with Project Examples

2. Simple Example of Using SQL

Print records in Boats Table that Boat's color is red



Result



Tutorial with Project Examples

3. Simple tests

Names of sailors who have booked red boats (1)

Tutorial with Project Examples

3. Simple tests

Names of sailors who have booked red boats (2)

Tutorial with Project Examples

3. Simple tests

Names of sailors who have booked boats within 2019 (1)

Tutorial with Project Examples

3. Simple tests

Names of sailors who have booked boats within 2019 (2)

Tutorial with Project Examples

3. Simple tests

Boat names booked by at least 2 different sailors

Project 2

I. Tutorial with Project Examples

II. Assignment

Assignment

1. Assignment Content

1. Use **SQL** queries on university database.

2. Write **SQL** queries that answer the questions below (one query per question) and run them on the Microsoft ACCESS Database System using its SQL interpreter.

Assignment

1. Assignment Content

3. Questions

1. **Print the names of professors who work in departments that have fewer than 50 PhD students.**
2. **Print the name(s) of student(s) with the lowest gpa.**
3. **For each Computer Sciences class, print the cno, sectno, and the average gpa of the students enrolled in the class.**
4. **Print the course names, course numbers and section numbers of all classes with more than six students enrolled in them.**
5. **Print the name(s) and sid(s) of the student(s) enrolled in the most classes.**
6. **Print the names of departments that have one or more majors who are under 18 years old.**

Assignment

1. Assignment Content

3. Questions

7. Print the names and majors of students who are taking one of the College Geometry courses. (Hint: You'll need to use the "like" predicate and the string matching character in your query).

8. For those departments that have no majors taking a College Geometry course, print the department name and the number of PhD students in the department.

9. Print the names of students who are taking both a Computer Sciences course and a Mathematics course.

10. Print the age difference between the oldest and youngest Computer Sciences major(s).

11. For each department that has one or more majors with a GPA under 1.0, print the name of the department and the average GPA of its majors.

12. Print the ids, names, and GPAs of the students who are currently taking all of the Civil Engineering courses.

Assignment

1. Assignment Content

4. Schema and Sample Data

- **The schema of the database is provided below (keys are red, field types are omitted):**
  - **student(sid, sname, sex, age, year, gpa)**
  - **dept(dname, numphds)**
  - **prof(pname, dname)**
  - **course(cno, cname, dname)**
  - **major(dname, sid)**
  - **section(dname, cno, sectno, pname)**
  - **enroll(sid, grade, dname, cno, sectno)**

Assignment

1. Assignment Content

5. Data Files

- **Download the following data files from WeChat Group(The imports you made in Project 1).**
  - **course.txt**
  - **dept.txt**
  - **enroll.txt**
  - **major.txt**
  - **prof.txt**
  - **section.txt**
  - **student.txt**

It is recommended to use the original student.txt for better communication and comparison of correctness among classmates.

Assignment

2. A Few Points to Note

- The query answers must not contain duplicates, and the SQL keyword **distinct** should only be used when necessary.
- The SQL interpreter in ACCESS is not quite the same as the one described in the textbook. If the query you write is not accepted by ACCESS (usually it gives you some strange errors), try different ways until you get one that works with ACCESS.
- For this assignment, creation of temporary tables is not allowed, i.e., for each question you have to write exactly one SQL statement.

Assignment

2. A Few Points to Note

- This is an **individual assignment** – no group submissions are allowed.
- Hand in **an ACCESS database** that contains the answers to the 12 questions. The database should **contain 12 queries**, named as follows:

  Query1：

  Query2：

  …

  Query12：
- Test the function of index with query related with student table.
- Hand in a report which outlines your answers and the process by which you arrived at them.

How to improve your SQL capability?

I. nowcoder

II. LeetCode

III. AI